

DISEASES OF INDIAN ELEPHANTS : AN OVERVIEW

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Elephants are largest of living warm blooded land mammals belonging to order *Proboscidea*. African (*Loxodonta africana*) and Asian (*Elephas maximus*) are two species of elephants alive today.

I. Digestive system

- The elephants are simple stomach animals and alimentary canal is similar to horse. Elephants lack gall bladder, however bile duct is large and sacculated as it leaves liver.
- Normal rectal temperature is 35.6°C or 96°F (Yathiraj *et al.*, 1992)
- Dental formula: Upper jaw: 2 (1 I, 0 C, 6 M), Lower jaw: 2 (1 I, 0 C, 6 M)
- Modified incisors are composed of dentine (ivory).
- Normally food takes 24-50 hours through elephant's digestive tract. Normal diet should include grass hay and commercial equine grains or pellets. 10-25% of grass hay should be substituted with legume hay along with vegetable and fruits.

1. Tusk problem: Broken tusks are common problem in elephants. In many cases, tip of the tusk breaks off, hence exposing the pulp cavity. Metal caps can be used to plug the cavity. But if severe infection occurs, removal of tusk is necessary. Tusks grown out in abnormal directions can be corrected by employing orthodontic procedures. Infection of tusk sulcus involves the removal of the foreign bodies or necrosed bone fragments and regular flushing with the antibiotic or antiseptic solutions (Ranganath *et al.*, 1993).

2. Colic: The most common cause of colic in elephants is change of diet or over eating. Feed material that usually cause colic in elephants include corn stalk, watermelon and clover hay. Signs of colic in elephants are similar to that seen in equines. Frequent groaning is an indication of pain. Abdominal pain can lead to abnormal behaviour, restlessness, frequent lying down and getting up, placing trunk in mouth, biting the tip of the trunk, or assume abnormal postures like crossing the rear limbs. Treatment of colic includes copious soapy lukewarm water enema repeated until the elephant shows the signs of relief.

Complete Intestinal blockade caused by ingestion of foreign materials such as gravel, is accompanied by severe colic pain. Management of intestinal blockade includes repeated soapy water enema along with use of lubricants such as mineral oil orally and other supportive therapy like intravenous fluid therapy and oral electrolytes and rehydrating fluids.

3. Indigestion: Recent transport, poor feeding, changes of feed and climatic variations are the circumstances, which lead the elephants especially elephant calves to indigestion. Clinical symptoms include listlessness, muddy colour of the conjunctiva, passing small quantities of urine with dark coloured urine and mild abdominal pain and grunt and fever (Magesh and Dhanapalan, 2001). Management of simple indigestion includes administration of liver extracts, vitamin supplementation both orally and parentally, yeast and lactobacillus etc.

4. Diarrhoea: Most common causes are:

(i) **Salmonellosis:** It should be considered as a part of the differential diagnosis for severe diarrhoea in elephants. Signs of salmonellosis in elephants include profuse watery diarrhea containing blood and mucus, loss of appetite, weakness, fever and dehydration. Faecal culture or serological investigation helps in diagnosis. *Salmonella* bacilli are gram negative and non-lactose fermentors.

Treatment includes correction of fluid and electrolyte balance along with antibiotic therapy. Usually sulphonamides, tetracyclines, chloramphenicol are drugs of choice at the dose rate as recommended in horses. The amount of intravenous fluid for adult elephants can reach up to 100-150 litres per day. Oral electrolyte and rehydrating fluids can also be used and elephants should be encouraged to take cooked rice and drink plenty of fluids.

(ii) **Enterotoxaemia:** Young animals are more susceptible to clostridium enterotoxaemia. Food contamination is an important source of infection. Signs include diarrhoea, listlessness, anorexia and death. Diagnosis can be made from the culture and isolation of the organisms from the infected secretions and excretions.

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(iii). **Colibacillosis:** Usually occurs in newborn that does not receive colostrum. Signs of colibacillosis in elephant calves are similar to that seen in bovine calves. Treatment includes antibacterial drugs and fluid therapy.

(iv). **Rinderpest:** It has been reported in Asian elephants. Signs include elevated rectal temperature, profuse ocular discharge, erosions and ulcerations on mucosa of oral and vagina. Diarrhoea appears 2-4 days after the appearance of oral and vaginal lesions. Leukopenia occurs initially but is quickly followed by leukocytosis.

5. Diseases of liver: Hydatid cysts, cholelithiasis, cholangitis and hepatitis have been reported in elephants. Jaundice is a most common sign of liver disease. Biliary stones are usually associated with infestation of nematodes of biliary ducts like *Grammocephalus clethrotus*. Flukes like *Fasciola jacksoni*, they usually lead to anaemia and hypoproteinaemia (Singh *et al.*, 1984).

6. Choke: It has been reported in number of Asian elephants. The affected elephants are unable to swallow saliva and it drools from the mouth. A mass will be palpable or observable under the skin of neck where the obstruction is lodged in the oesophagus. If the obstruction does not dissolve or pass of its own, then stomach tube can be passed. If medicinal therapy fails the surgical intervention should be undertaken.

7. Foot and Mouth disease (FMD): FMD is a disease of great economic importance affecting the cloven footed animals. Various workers have reported the occurrence of this disease in elephants in India (Pyakural *et al.*, 1976; Rahman *et al.*, 1987; Chakroborty and Majumder, 1990; Sharma *et al.*, 1994). Sharma *et al.* (1994) described a case in baby elephant showing symptoms of high rise of body temperature (105° F), oedematous skin around the foot and nail, vesicular lesions on the trunk and tongue. Due to acute pain in the feet it was unable to move. Treatment includes washing the lesions with Candy lotion (1:10,000 solution of KMNO₄) along with broad-spectrum antibiotics and anti-inflammatory drugs. Isolation and confinement of the affected animal is essential to prevent further spread of the infection to the susceptible population.

8. Anthrax: Anthrax is caused by *Bacillus anthracis* and is epizootic in Indian elephants (Bapu, 1936-37). Out breaks of anthrax is reported in northeastern region of the country (Gupta 1928; Ahmed, 2001). Major signs are subcutaneous swelling, bloody diarrhoea, high fever, colic, hot painful swellings and sudden death (Gupta 1928).

II. Respiratory System

- Respiratory system begins with elongation and joining of upper lip and nose.
- Trachea is 5-7 cm in diameter and 30 cm in length; tracheal rings are incomplete dorsally. Elephant's lung fail to collapse like that of horse.
- Pleural cavity is absent. In sternal recumbency elephants show great respiratory distress including apnoea.
- Normal respiration rate in Indian elephants is 8-10 per minute (Yathiraj *et al.*, 1992).

Common diseases encountered are:

- 1. Tuberculosis:** It is a chronic infectious and contagious disease of great public health importance. Pulmonary T.B. is reported by Narayanan (1925) and Vijayan *et al.* (1992). Main clinical signs of T.B. in elephants are chronic weight loss despite adequate nutrition, lethargy, dyspnoea, coughing and chronic nasal discharge. Diagnosis can be made by intradermal tuberculin test (Bhat *et al.*, 1999). Further diagnosis includes tracheal wash and their culture examination. Treatment with drugs such as isoniazid, rifampin and streptomycin has been reported with variable success.
- 2. Tracheitis and bronchitis:** Clinical signs include hyperaemia, purulent exudate, drooping of the trunk, cough, laboured respiration, fever, anorexia and increased respiration rate.
- 3. Coryza like syndrome:** It is a viral infection affecting elephants. This syndrome occurs in the form of cold or catarrh and important clinical signs are excessive serous nasal discharge, watery discharge from eyes, decreased activity and listlessness. Healthy elephants should be able to recover from the disease without treatment. Supportive care and antibiotic therapy should be indicated.
- 4. Pasteurellosis:** Rare reports are available regarding this disease. Shrivastav and Chaturvedi (2000) reported a case of pasteurellosis in an elephant showing predominantly respiratory signs with high fever, anorexia, recumbency and death. Postmortem examination revealed septicaemia, pulmonary oedema, pneumonia and haemorrhagic endocarditis. Cultural and biochemical characterization of the isolate confirmed the etiological agent (*P. multocida*).

III. Cardiovascular system

- The heart of elephants is unique. It has double apex because of increased development of right

ventricle. Blood collection from the elephants is easiest from the large auricular veins, when animal is in lateral recumbency. It is made easier by swabbing the back of ear with luke warm water prior to collection to cause the veins to stand out.

- Elephants can be bled from the cephalic veins located on the proximal medial fore limb while standing and saphenous vein on the lower medial aspect of the hind limb while animal is standing or in lateral recumbency.
- Heart rate in elephants ranges from 34-46 per minute. It can rise to 60 per minute with strenuous exercise or colic.
- The electrocardiogram recording are easily made out at low voltage and are best recorded with lead I. Blood volume was estimated 3.5% of the body weight.
- The elephant's erythrocyte is largest of any mammal with the diameter 8.8-10.6 micrometre.
- The percentage of neutrophils and lymphocytes in Asian elephants is 27-45 and 52-72, respectively, while the haemoglobin ranged from 8.4-12.6 g/dl. Total leukocyte count ranged between 13,000-20,000 per cumm (Yathiraj *et al.*, 1992; Kumar *et al.*, 2004).

Auscultation of heart sounds is difficult but pulse rate is recorded by palpation of arteries on the ear pinna.

1. **Anaemia:** Clinical signs of anaemia include anorexia, pallor and depression. Iron deficiency due to inadequate dietary intake is one of the major cause of anaemia. Ferrous dextran @ 0.5 mg/kg body weight brings dramatic improvement within several days. Other important causes are blood parasites (haemoprotozoa). *Trypanosoma brucei* is reported as a causative agent of Trypanosomiasis in African elephants and *Trypanosoma evansi* in Asian elephants. Clinical signs include acute clinical onset, anaemia, subcutaneous oedema and emaciation. Treatment includes use of Quinapryamine sulphate in elephants. Babesiosis has been described within erythrocytes of African elephants. It is responsible for haemolytic anaemia. Treatment should include blood transfusion and Quinapryamine sulphate.

IV. Disease of Skin

- Sweat glands are present throughout the skin.
 - In elephants, skin is extremely thickened leading to general classification of pachyderm or thick skin.
1. **Pododermatitis:** It is very much common in captive elephants due to inadequate skin care. Optimal skin care includes daily bathing or hosing along with brick scrubbing with mildly abrasive materials such as pumice stones or stiff brush. Elephants that are not bathed or hosed regularly

will develop dirty, dry, warty skin along with pruritis (Jayathangaraj *et al.*, 2001).

2. **Ulcerations:** Common in debilitated animals and animals which are kept under poor sanitary conditions. Treatment consists of application of antibiotic ointments such as nitrofurazone dressing.
3. **Subcutaneous oedema:** Large pockets of oedema occur usually at dependent parts under conditions of great anxiety or stress. Small pockets of fluid spontaneously get absorbed. Large accumulated fluids required surgical drainage. Diuretics can be tried along with surgery. Subcutaneous oedema can also occur in gastrointestinal strongylosis (Kumar *et al.*, 2003).
4. **Cracked heel:** Diseases of the feet of the elephants are the greatest of all the medical problems. Domesticated elephants or that maintained in the zoological parks are chained in their legs and they are confined to stand or rest on bedding which is soaked with urine or faeces of the animal, thus it forms a paradise of various infecting agents. Cracked heel is a chronic disease that usually occurs about the junction of the horny sole and the skin. Poor sanitation and keeping the animal in wet conditions aggravates the condition. Symptoms vary with the degree of the disease. Chatterjee (1984) described lesions of cracked heel caused by *Stephanofilaria assamensis* in Asiatic elephants.

Lesions on the heel: At the junction of the skin and the horny sole, cracks were observed. Skin and the sole was separated and was bridged by serrated tissue. Lesion is usually moist without any oozing. Hind limbs are more involved. In advanced cases extensive pododermatitis of the sole was observed and animal showed limping during walk.

Lesions above the heel: About 3-5 cm above the heel multiple nodules measuring 5-10 mm in diameter were noticed. Smaller one is usually hard with intact skin as compared to larger one, which is usually soft and discharge thick sticky and creamy pus. These are painful on pressure. There is achromasia around the nodules

V. Urinary System:

- Kidneys are multilobulated.
 - Normal urine is straw coloured range from clear to turbid. The average specific gravity of urine is 1.019 (range 1.010 - 1.035). Average volume per void 5-11 litres (total per day 50 litres).
1. Cystic kidneys have been reported in elephants.
 2. Urolithiasis and Pyelonephritis: Urolithiasis occurs in all the species of the animals irrespective of age and sex. More commonly it

occurs in animals which are grazed on pasture containing high levels of silica, which may be as high as 6% (Bailey, 1981). There are sporadic reports of this disease in Indian elephants (Varma *et al.*, 2001). Diets low in vitamin A has also been reported as a cause of urolithiasis in animals. Varma *et al.* (2001) reported a rare case of chronic urolithiasis combined with acute pyelonephritis in a female Asian elephant. Clinical symptoms included anorexia, dullness, pyrexia, and slightly arched back posture with yellowish white discharge in urine. Postmortem examination revealed congestion and small abscesses in the renal cortex and severe congestion of the pelvis along with calculi.

VI. Nervous System Diseases

1. Rabies: A case of paralytic rabies in elephant cow following dog bite was reported by elephants (Ramiah, 1932-33). Animal developed sudden lameness, hemiplegia and complete paresis. Progressive impairment of appetite and fever (98.4°F to 99.8°F) was recorded leading to death on 9th day of illness. Gopal and Rao (1984) also reported rabies in young elephant male calf. Animal showed symptoms of involuntary defaecation, weak hind limb and tail reflexes.
2. Heat stroke: Elephants exposed to high ambient temperatures and direct sun light is susceptible to heat strokes, which will usually terminate in the death of the animal.

VII. Parasitic diseases

Parasitic infection is a common problem in Asiatic elephants. Non-specific clinical symptoms of parasitism include depression, inappetence, soil-eating behaviour, loss of body condition, lachrymation, anaemia, subcutaneous oedema, alternate profuse diarrhoea and constipation.

External parasites

(i). *Haematomyzus elephantis* (elephant lice) has been reported in Asian elephants on both sides of ear flaps, neck and ventral aspect of the body (Arora and Prasad, 1996). Pruritus and dryness of the skin are usually main clinical signs besides the loss of appetite. The reasons for lice infestation could be attributable to lack of adequate grooming and/or not having adequate opportunities to animal for taking bath.

Intestinal Parasites: Common nematodes of small intestine are *Murshidia*, *Quilonia*, and *Amira species*. Signs of intestinal parasitism include gastroenteritis, loss of body weight, anaemia and colic. Either alone or concurrent infection of strongylosis and amphistomiasis has been reported in elephants (Rao *et al.*, 1992; Singh *et al.*, 2001; Kumar *et al.*, 2003) showing symptoms of diarrhoea,

subcutaneous oedema and soil eating behaviour. Fenbendazole @ 5.0 mg/kg body weight either as a single dose (Roy and Majumdar, 1988; Rao *et al.*, 1992; Raman *et al.*, 2000) or in two divided doses (Kumar *et al.*, 2003) has been reported with 100% success against strongylosis in elephants. Islam *et al.*, (1999) and Muraleedharan and Keshavaprasad (2001) reported *Pseudodiscus collinsi* and *Gastrodiscus secundus* amphistomes from caecum of Asian elephants. Modi (2001) recorded prevalence of helminth parasites in elephants of Kahna of which *Stroglyoides* was highly prevalent infestation followed by *Ascaris*, *Schistosoma*, *Amphistomes*, *Oesophagostomum*, *Fasciola*, *Uncinaria* etc.

VIII. Surgical Diseases

1. **Granuloma:** Chronic wounds resulting from repeated trauma, especially on the limbs, which become either refractory to conventional treatment or due to indiscriminate use of therapeutic regimens may exacerbate the lesions and precipitate hyperplastic changes within the tissues leading to formation of granulomas. Surgical enucleation of the granulomatous lesion *en mass* is recommended procedure in the treatment (Angelo *et al.*, 1976).
2. **Abscess:** They are mostly undetected for long periods of time due to thick skin. They occur as a result of contusions, wounds, parasites etc. They are diagnosed as initially hot hard swellings and later on become cold fluctuating swellings underneath the skin. Treatment includes surgical exploration, complete drainage followed by flushing with antibiotics and antiseptic solutions. Flushing with nitrofurazone (2 parts) and dimethylsulphoxide (1 part) has good results.
3. **Farragall:** Farragall resembles saddle gall in horses. It is the chronic inflammatory condition leading to wound in the neck of elephant, where the mahout used to tie the seat. The weight of the mahout and constant movement of the skin of the elephant during walking aggravates the condition. The surgery is indicated extirpate the fibrotic mass under general anesthesia (Sarma *et al.*, 2001).
4. **Pharyngeal Fistula:** Chandrapuria and Shrivastava (2001) reported surgical treatment of pharyngeal fistula.
5. **Prepuccial Diverticulum:** Sarma and Dutta (1997) reported a case of prepuccial diverticulum and its surgical treatment in an elephant.
6. **Umbilical hernia:** Singh (1971) described successful surgical repair of umbilical hernia.

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